

# AI-Powered Enterprise E-Commerce Platform

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## Non-Functional Requirements Specification (NFRS)

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### Document Information

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| Item          | Description                               |
|---------------|-------------------------------------------|
| Project Name  | AI-Powered Enterprise E-Commerce Platform |
| Document Type | Non-Functional Requirements Specification |
| Version       | 1.0                                       |
| Prepared By   | Project Team                              |
| Status        | Draft                                     |

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## 1. Introduction

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Non-Functional Requirements (NFRs) describe the quality characteristics and operational constraints of a software system.

Unlike Functional Requirements, which describe **what the system should do**, Non-Functional Requirements describe **how well the system should perform those functions**.

Examples include:

- Performance
- Security
- Scalability

- Reliability
- Availability
- Maintainability
- Usability

These requirements are essential for developing enterprise-level systems.

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## 2. Purpose

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The purpose of this document is to define the quality requirements of the AI-Powered Enterprise E-Commerce Platform.

This document helps:

- improve system quality,
  - reduce operational risks,
  - guide system architecture,
  - establish measurable standards,
  - support testing and deployment.
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## 3. Importance of Non-Functional Requirements

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A system may have all required features but still fail if:

- it is slow,
- it crashes frequently,
- it is difficult to maintain,
- it is insecure,
- or users find it difficult to use.

Therefore, NFRs are critical to the success of software systems.

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## 4. Non-Functional Requirement Categories

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The following categories have been identified.

1. Performance Requirements
2. Scalability Requirements
3. Availability Requirements
4. Reliability Requirements
5. Security Requirements
6. Maintainability Requirements
7. Usability Requirements
8. Compatibility Requirements

9. Portability Requirements
  10. Monitoring Requirements
  11. Logging Requirements
  12. Backup and Recovery Requirements
  13. Compliance Requirements
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## 5. Detailed Non-Functional Requirements

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### 5.1 Performance Requirements

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Performance refers to how quickly and efficiently the system performs its operations.

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#### NFR-001 Response Time

The system should respond to normal user requests within:

- Product Search → less than 3 seconds
- Login → less than 2 seconds
- Product Recommendation → less than 5 seconds

Priority:

Critical

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#### NFR-002 Concurrent Users

The system should support multiple users simultaneously.

Target:

- At least 1,000 concurrent users.

Priority:

High

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#### NFR-003 Database Performance

Database queries should execute efficiently.

Target:

- Standard queries should complete within 2 seconds.

Priority:

High

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## NFR-004 AI Processing Performance

Recommendation and forecasting services should produce results within acceptable time limits.

Priority:

Medium

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## 5.2 Scalability Requirements

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Scalability refers to the system's ability to handle increasing workloads.

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### NFR-005 Horizontal Scalability

The system should allow additional servers to be added when traffic increases.

Priority:

High

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### NFR-006 Database Scalability

The database should support increasing amounts of data.

Expected Growth:

- Millions of products
- Millions of customers
- Millions of transactions

Priority:

High

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## 5.3 Availability Requirements

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Availability refers to how often the system remains operational.

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### NFR-007 System Availability

Target Availability:

99.9%

The system should remain operational except during scheduled maintenance.

Priority:

Critical

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## NFR-008 Maintenance Downtime

Maintenance activities should minimize service interruption.

Priority:

Medium

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## 5.4 Reliability Requirements

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Reliability refers to the ability of the system to operate consistently.

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### NFR-009 Failure Rate

The system should experience minimal failures.

Priority:

Critical

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### NFR-010 Data Integrity

Transactions should not result in data corruption.

Priority:

Critical

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### NFR-011 Error Recovery

The system should recover gracefully from failures.

Priority:

High

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## 5.5 Security Requirements

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Security is one of the most important aspects of enterprise applications.

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## NFR-012 Authentication Security

All users should be authenticated before accessing protected resources.

Priority:

Critical

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## NFR-013 Password Security

Passwords should:

- be encrypted,
- never be stored in plain text.

Priority:

Critical

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## NFR-014 Authorization

The system should implement Role-Based Access Control (RBAC).

Priority:

Critical

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## NFR-015 Secure Communication

Sensitive information should be transmitted securely.

Examples:

- HTTPS
- TLS

Priority:

Critical

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## NFR-016 Session Security

Sessions should automatically expire after inactivity.

Priority:

High

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## NFR-017 Protection Against Common Attacks

The system should protect against:

- SQL Injection
- Cross Site Scripting (XSS)
- Cross Site Request Forgery (CSRF)
- Broken Authentication
- Unauthorized Access

Priority:

Critical

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## NFR-018 Fraud Protection

The system should identify suspicious activities.

Priority:

Critical

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## 5.6 Maintainability Requirements

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Maintainability refers to how easily the system can be modified.

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### NFR-019 Modular Architecture

The system should follow modular design principles.

Priority:

High

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### NFR-020 Documentation

System documentation should be maintained.

Priority:

High

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### NFR-021 Coding Standards

Developers should follow coding standards and best practices.

Priority:

Medium

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## NFR-022 API Documentation

All APIs should be properly documented.

Priority:

Medium

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## 5.7 Usability Requirements

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Usability refers to how easy the system is to use.

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### NFR-023 User-Friendly Interface

The system should provide intuitive interfaces.

Priority:

High

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### NFR-024 Learnability

New users should be able to understand basic operations quickly.

Priority:

Medium

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### NFR-025 Accessibility

Interfaces should consider accessibility principles.

Priority:

Medium

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## 5.8 Compatibility Requirements

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### NFR-026 Browser Compatibility

The system should work on:

- Google Chrome
- Microsoft Edge



- Firefox

Priority:

Medium

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## NFR-027 Device Compatibility

The system should support:

- Desktop
- Laptop
- Tablet
- Mobile devices

Priority:

Medium

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## 5.9 Portability Requirements

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### NFR-028 Platform Independence

The system should operate on different operating systems.

Examples:

- Windows
- Linux
- macOS

Priority:

Medium

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### NFR-029 Container Support

The system should support containerized deployment.

Examples:

- Docker
- Kubernetes

Priority:

Medium

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## 5.10 Monitoring Requirements

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Monitoring is extremely important in enterprise systems.

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### NFR-030 Health Monitoring

The system should provide health monitoring information.

Priority:

High

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### NFR-031 Performance Monitoring

The system should monitor:

- CPU usage
- Memory usage
- Response times

Priority:

Medium

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## 5.11 Logging Requirements

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### NFR-032 Activity Logging

Important activities should be logged.

Examples:

- Login attempts
- Product updates
- Transactions

Priority:

High

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### NFR-033 Error Logging

Errors should be logged for debugging.

Priority:

High

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## NFR-034 Audit Logs

Administrative activities should be auditable.

Priority:

High

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## 5.12 Backup and Recovery Requirements

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### NFR-035 Database Backup

The system should regularly back up data.

Priority:

Critical

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### NFR-036 Disaster Recovery

Recovery procedures should exist for major failures.

Priority:

High

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### NFR-037 Recovery Time Objective (RTO)

Target:

Less than 4 hours.

Priority:

Medium

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### NFR-038 Recovery Point Objective (RPO)

Maximum acceptable data loss:

Less than 30 minutes.

Priority:

Medium

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## 5.13 Compliance Requirements

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### NFR-039 Privacy Protection

Customer information should be protected.

Priority:

Critical

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### NFR-040 Data Retention Policies

The system should define how long data is stored.

Priority:

Medium

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### NFR-041 Legal Compliance

The system should comply with applicable regulations.

Examples:

- GDPR principles
- Data Protection Laws

Priority:

Medium

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## 6. Quality Attributes

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The following quality attributes are expected.

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### Performance

Fast response times.

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### Security

Protection against threats.

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## Reliability

Consistent operation.

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## Scalability

Ability to support growth.

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## Maintainability

Easy future modifications.

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## Availability

Continuous operation.

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## Usability

Ease of use.

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# 7. Acceptance Criteria

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The system will be considered acceptable if:

### Performance

- Response times satisfy requirements.

### Availability

- Availability reaches 99.9%.

### Security

- No major security vulnerabilities.

### Reliability

- Minimal critical failures.

### Maintainability

- New modules can be added with minimal changes.
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## 8. Non-Functional Requirement Traceability Matrix

| Business Goal                 | Related NFR        |
|-------------------------------|--------------------|
| Improve customer satisfaction | NFR-001, NFR-023   |
| Improve scalability           | NFR-005, NFR-006   |
| Improve security              | NFR-012 to NFR-018 |
| Improve reliability           | NFR-009 to NFR-011 |
| Improve maintainability       | NFR-019 to NFR-022 |

## 9. Conclusion

The Non-Functional Requirements defined in this document ensure that the AI-Powered Enterprise E-Commerce Platform is not only feature-rich but also secure, scalable, reliable, maintainable, and suitable for enterprise environments.

These requirements provide the foundation for future:

- System Architecture
- Database Design
- Deployment Planning
- DevOps Activities
- Testing Strategies
- Production Readiness

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